Ex.No: 04
Date: 03/08/2026 Setting up a Local Area Network

Aim:

To design, set up and configure a Local Area Network connecting four computers through a network switch, assign static IP address to all devices, Verify connectivity using the ping command and demonstrate resource data sharing across the network.

Diagram:

LAN was implement using STAR Topology

Implementation:

Step 1: place the Device.

Open the CISCO packet Tracer.

From the device category panel, select End Devia & drag 4 PCs onto the workspace.

Select Switches and drag 1 switch onto the workspace.

Step 2: Cable the Device

Click Connections → Select copper-Straight Through cable.

Click each PC, then click the switch and select an available Fastethernet port to complete to each connection.

Repeat for all four PCs. Green link lights confirm a Successful physical connection.

Step 3: Assign IP Address to PCs.

Click a PC → Desktop tab → IP configuration

Enter the static IP Address and Subnet Mask as per the IP table for each PC.

Step 4: Configure the Switch

Click a Switch → CLI Tab

Enter the following commands to assign a management IP.

commands:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit.
```

Step 5: Verify Connectivity Using ping.

Click a PC → Desktop tab → Command prompt

use the ping command to test connectivity to the other PC,

Succesful "Reply from..." message confirm connectivity.

Step 6: Demonstrate Data Sharing

Add a Server device via its and connect it to the Switch

Assign it an IP address via its Desktop → IP configuration Tab.

Open the Service tab and enable the HTTP Service.

From another PC, open the Web Browser & enter the Server's IP address to load the page, confirming Succesful Data Transfer across the LAN.

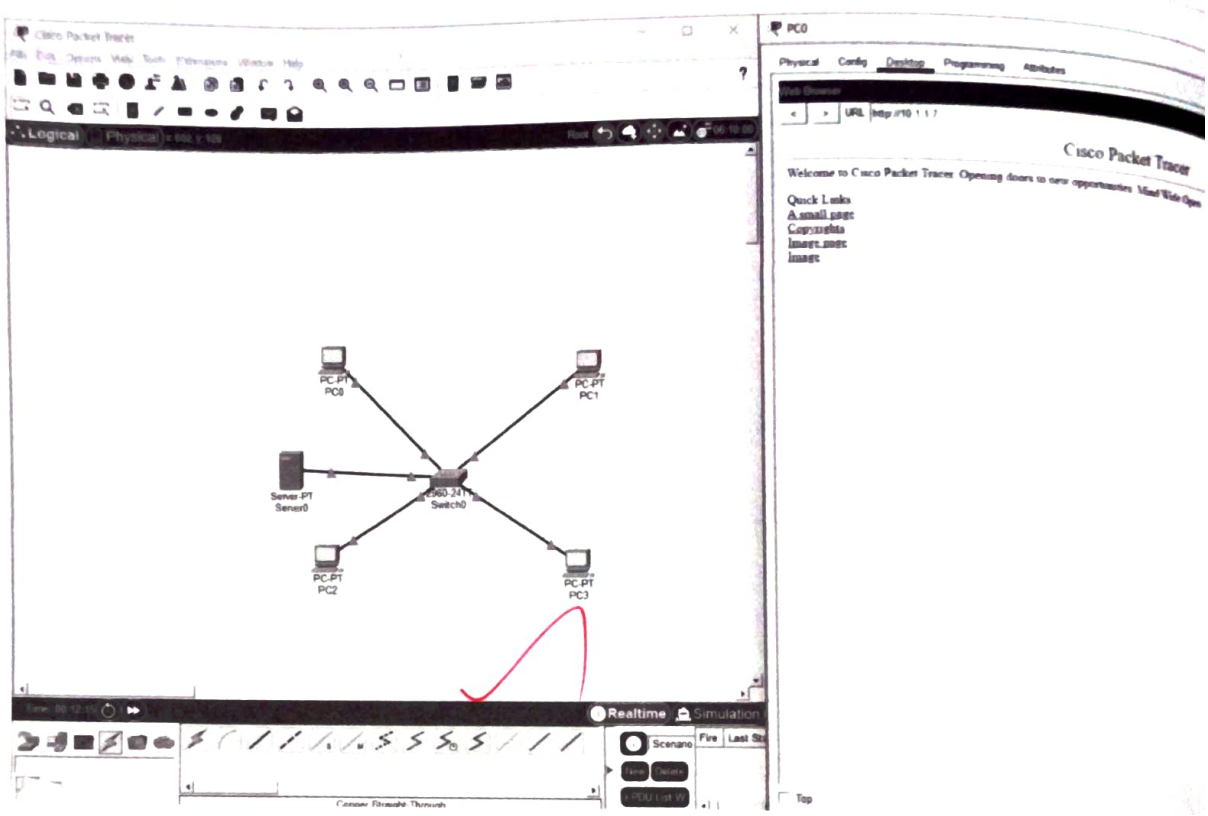
Step 7: Observe Packet flow in Simulation Work

Switch to Simulation mode

Send a ping or web request b/w two devices & use the play controls to step through the packet as it travels hop-by-hop through the Switch, observing MAC-address-based forwarding.

7. Observation:

All four PCs were successfully connected to the Switch in a star topology & assigned unique static IP Address within the same subnet. Pings tests b/w all devices returned successfull replies.




Result:

Setting Up a Local Area Network (LAN) was learned & Executed Successfully.